Lab 1:

2. Implement a simple calculator program.

Lab Tasks:

1. Accept numeric input.
2. Perform addition, subtraction, and multiplication using integer types.
3. Perform addition, subtraction, and multiplication using floating-point types.
4. Use if statements and loops to validate the input (reject non-numeric or out-of-range values) and handle different scenarios. (Experiment 1)

```go
package main

import "fmt"

func main() {
    var a, b int
    var x, y float64

    // Integer input validation
    for {
        fmt.Print("Enter first integer: ")
        _, err := fmt.Scan(&a)

        if err == nil && a >= -1000000 && a <= 1000000 {
            break
        }

        fmt.Println("Invalid input! Enter a valid integer.")
        var discard string
        fmt.Scanln(&discard)
    }

    for {
        fmt.Print("Enter second integer: ")
        _, err := fmt.Scan(&b)

        if err == nil && b >= -1000000 && b <= 1000000 {
            break
        }

        fmt.Println("Invalid input! Enter a valid integer.")
        var discard string
        fmt.Scanln(&discard)
    }

    // Integer operations
    fmt.Println("\nInteger Operations:")
    fmt.Println("Addition:", a+b)
    fmt.Println("Subtraction:", a-b)
    fmt.Println("Multiplication:", a*b)
```

```go
func main() {
    fmt.Println("Multiplication:", a*b)

    // Floating-point input validation
    for {
        fmt.Print("\nEnter first float number: ")
        _, err := fmt.Scan(&x)

        if err == nil && x >= -1000000 && x <= 1000000 {
            break
        }

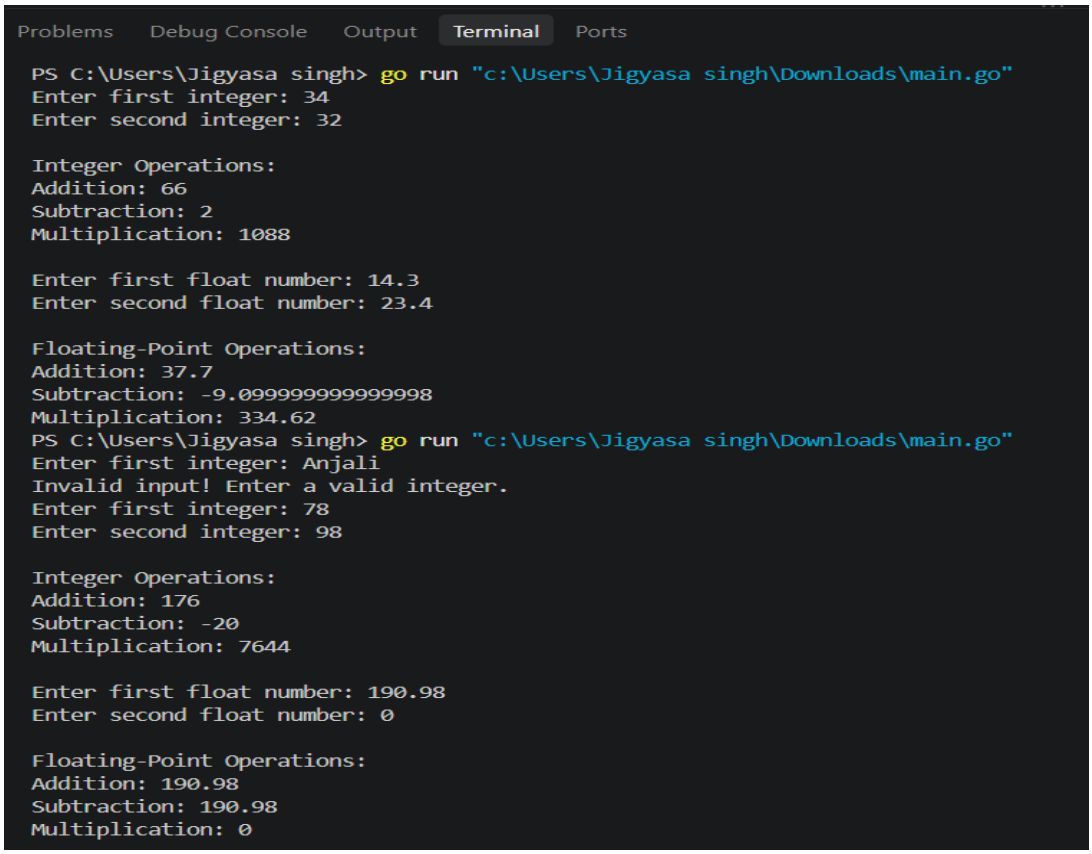
        fmt.Println("Invalid input! Enter a valid float number.")
        var discard string
        fmt.Scanln(&discard)
    }

    for {
        fmt.Print("Enter second float number: ")
        _, err := fmt.Scan(&y)

        if err == nil && y >= -1000000 && y <= 1000000 {
            break
        }

        fmt.Println("Invalid input! Enter a valid float number.")
        var discard string
        fmt.Scanln(&discard)
    }

    // Floating-point operations
    fmt.Println("\nFloating-Point Operations:")
    fmt.Println("Addition:", x+y)
    fmt.Println("Subtraction:", x-y)
    fmt.Println("Multiplication:", x*y)
}
```


Problems
Debug Console
Output
Terminal
Ports
PS C:\Users\Jigyasa singh> go run "c:\Users\Jigyasa singh\Downloads\main.go"
Enter first integer: 34
Enter second integer: 32
Integer Operations:
Addition: 66
Subtraction: 2
Multiplication: 1088
Enter first float number: 14.3
Enter second float number: 23.4
Floating-Point Operations:
Addition: 37.7
Subtraction: -9.099999999999998
Multiplication: 334.62
PS C:\Users\Jigyasa singh> go run "c:\Users\Jigyasa singh\Downloads\main.go"
Enter first integer: Anjali
Invalid input! Enter a valid integer.
Enter first integer: 78
Enter second integer: 98
Integer Operations:
Addition: 176
Subtraction: -20
Multiplication: 7644
Enter first float number: 190.98
Enter second float number: 0
Floating-Point Operations:
Addition: 190.98
Subtraction: 190.98
Multiplication: 0